Practical 6 Non-Clastic Sedimentary Rocks

(Limestones, Dolomites, Cherts ad Evaporites)

"Ah! Vanitas vanitatum! Which of us is happy is this world? Which of us has his desire? Or, having it, is satisfied? – Come, children, let us shut up the box and the puppets, for our play is played out."

William Makepeace Thackeray (1811-1863)

The class this week is devoted to the study of carbonate sediments and rocks with the emphasis on recognizing and interpreting grain types and fabrics as seen in hand specimen and thin section. We will also be looking at other sedimentary rocks, particularly the evaporites.

By the end of today's class you should:

- be able to recognize some of the more common grain types in limestones
- understand the Dunham classification of limestones and be able to apply it tentatively to hand specimens
- be able to interpret depositional mechanisms of common limestone types
- aware of the different types of carbonate mudstone and how to identify them
- able to estimate amounts and describe types of porosity in sedimentary rocks
- able to recognize dolomite in hand specimen
- able to recognize chert and the common evaporate minerals in hand specimen and understand their origin

A. LIMESTONES

L242 and R33

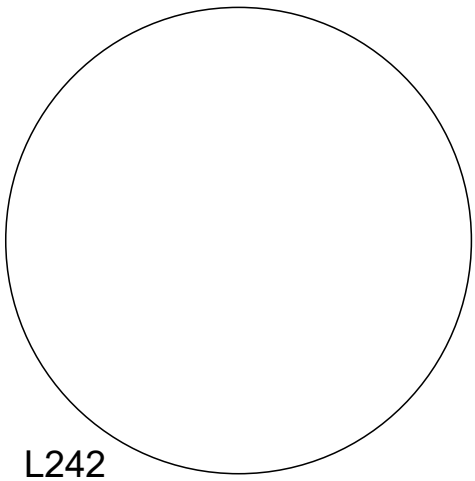
These are typical limestones from the Middle Jurassic of Britain. The principal grain type in both is the same and should be readily identifiable if you look at both samples with a hand lens and either sample in thin section.

These grains are …… and can be identified by

……

……

Make sure that you show on your thin section sketch the diagnostic internal structure you used to identify the grain type present.



L242
or R33

With limestones like L242 and R33 currents play a significant role in transport and deposition so grain size and textural features are important indicators of depositional conditions.

These rocks are both …… (limestone grain size classification),

but L242 has an average size of …… mm and R33 of …… mm.

Deciding on the correct name using the Dunham Classification is difficult without a thin section. It depends on recognizing grain or matrix support and estimating the amount of carbonate mud matrix present.

Examine a hand specimen and a thin section of either L242 or R33.

Both L242 and R33 are …… supported, so they must classify as

…… or …….....

Careful inspection in thin section of the spaces between the grains may reveal that it has sparite cement, or carbonate mud matrix, or areas of both, filling the pore spaces between the grains. Illustrate the key attributes of these pore filling materials on your thin section sketch.

Conclude the correct name for your specimen using the Dunham classification based on whether you observe the presence or absence of carbonate mud.

.....

R31 and L210

Many limestones are made up of the complete or fragmented remains of the hard parts of organisms. These are all called bioclasts. Look at R31 and L210 in thin section.

What is the composition of these bioclasts?

……

R31 and L210 would therefore be called (rock name) bioclastic limestones, in which the dominant organisms represented in these two samples are broken-

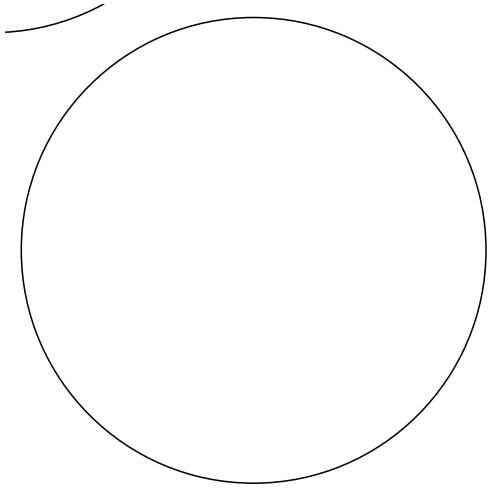
up …… .

Limestones of this type are particularly common in the Palaeozoic. R31 consists almost entirely of the stems and plates of these organisms and would

probably classify (Dunham) as a ……...

In L210 the fragments are supported by …… and the sample would

therefore classify as a …….



R31 or L210

L195

This is another bioclastic limestone. In this case the bioclasts are

(organism) ……. . There is a matrix of …….. so this

limestone would probably classify (Dunham) as a ……. Nearly all the bioclasts appear to be of the same organism. Are the shells complete or

fragmented? ……. What might these features and the rock texture suggest about the environment of deposition?

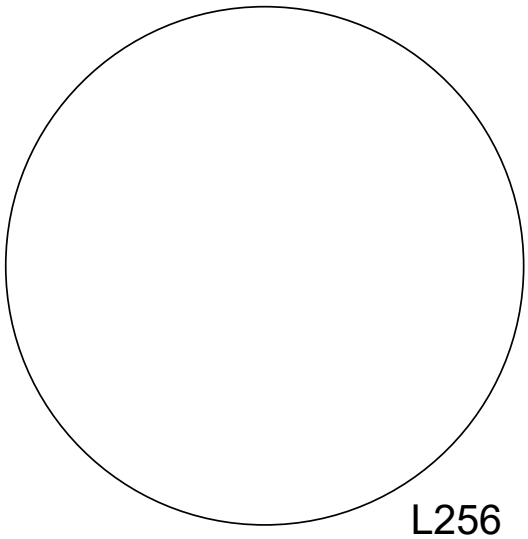
……

……

L256

Examine this rock carefully with a hand lens and in thin section. What do you think would be the most appropriate name for the grains in L256?

…….



L256

Annotate your sketch to show the diagnostic features you used to make this interpretation.

S129

Boundstones, where the rock is bound together in the depositional environment, are difficult to recognize in isolated hand specimens. They are usually biogenic rocks where larger calcareous organisms have constructed a rigid framework such as in a reef. S129 might be regarded as a boundstone.

It consists of …… (type of organism) with

…… between the …… .

B. CARBONATE MUDSTONES

L196 and R34 are both very fine-grained limestones and it would therefore be appropriate to classify them as carbonate mudstones, although they are very different in appearance. Describe and compare them (e.g., colour, hardness, density, porosity, composition, sedimentary strutcures) and account for the differences between them.

……

……

……

……

……

……

R34 is from one of the most well-known of all geological formations, the Upper Cretaceous Chalk. L196 is from the Carboniferous Limestone. What can you say about the likely depositional conditions of L196 and R34?

You will need to take into account what is known about the Chalk from electron microscope studies (see lecture notes).

L196 ……

……

R34 ……

……

One of the features of the Chalk sample which you should have described is its porosity. Using one of the cut blocks of Chalk note its dimensions

…… and thus work out its volume ……

Now use the balance to record its weight …… The density of calcite is 2.71 $g.cm^{-3}$, so you can work out what volume the block would occupy if it were

solid calcite with no pore spaces ……

Now you should be able to work out the porosity of your block ……

……

This should be quite a high figure for a rock. What do you think the permeability of the Chalk might be (qualitatively) and why?

……

Some fine-grained carbonate rocks are laminated on a mm-scale and are called **stromatolites**. These form from the sediment trapping and binding activities of cyano-bacterial mats. The lamination represents the original layering of fine sediment alternating with organic-rich mat layers, although in ancient examples the organic material has decayed. Sketch the lamination in L106 in the space below, and suggest why you think the rock has been interpreted as a stromatolite.

……

……

C. DOLOMITES

Dolomite can be difficult to distinguish from calcite especially where it is fine-grained. L207 is a partially dolomitized limestone. Look at it carefully in hand specimens and thin section. Which component is dolomite and how can it be identified in this sample both in hand specimen and in thin section?

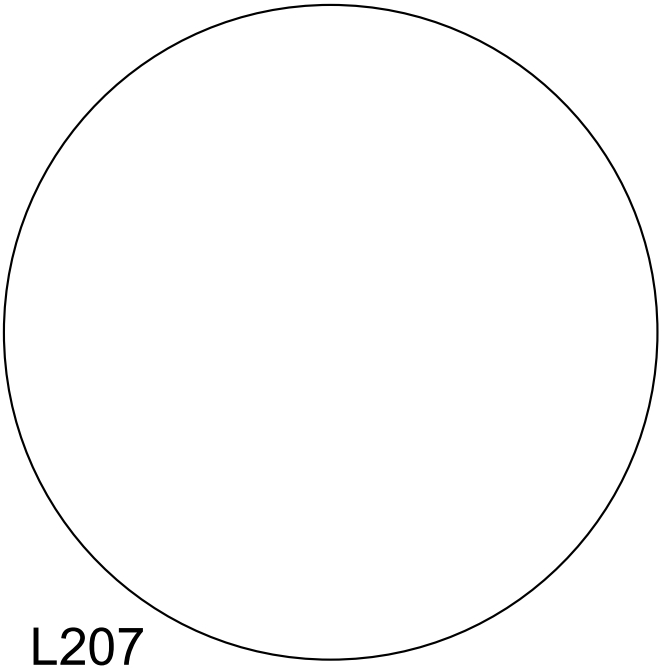
……..

……

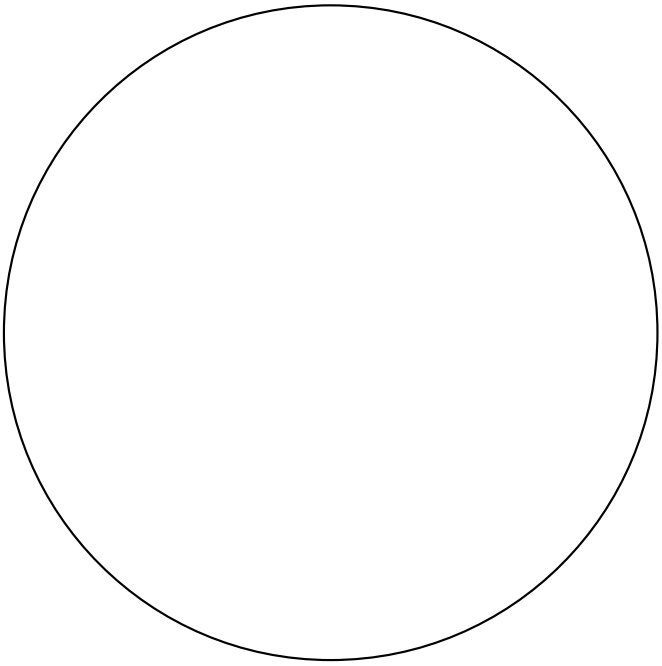
L173 is almost entirely composed of dolomite. Examine the hand specimen and thin section. How can this rock be identified as dolomite rather than, say,

a sandstone ……

or a limestone?…….



L207



L173

D. POROSITY

Hand specimen B10 is a porous limestone from the British Jurassic

Name the rock in as complete a way as possible based on the Dunham

textural name ……

Describe both primary and secondary porosity, outlining how the latter

developed……

……

E. CHERTS

S45 is a radiolarian chert, although you would need a high-powered microscope to see the radiolarians. It is thus a primary chert and features such as red colouration and banding are common. What is the cause of the red colour and what is it doing in a deep-water sediment?

……

……

R36 and R37 are replacement cherts. How can you tell that R36 is chert?

……

What was the nature of the original sediment before replacement?

……

How would you describe the porosity? ……

R37 is chert from the Chalk and commonly known as ……

What name do we give to the smooth fracture exhibited by most specimens of

R37? ……

F. EVAPORITES

After a little calcium carbonate, calcium sulphates are the first minerals to precipitate from evaporating saline water. Usually gypsum precipitates, but this may convert to anhydrite on burial. Two samples of sulphates are provided - FY3 and FY4. Identify them and summarize their distinguishing properties.

……

……

……

Specimen FY5 is halite or 'rock salt'. Summarize the properties by which you can recognize it (do not include taste – it isn't something you should try on lab specimens!)

……

……

Evaporites may leave traces of their former presence even though they are no longer there. Look at sample FY 6. Suggest an appropriate rock name

…… What evidence is there for the former presence

of evaporites?…….

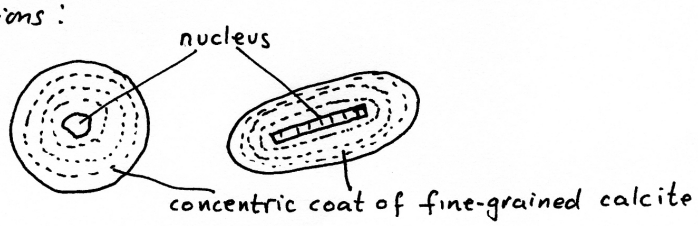
……

COMMON GRAIN TYPES IN LIMESTONES

Ooids

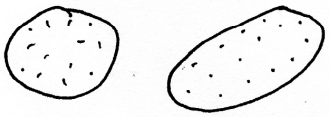
Spherical or ellipsoidal grains < 2mm in diameter with a 'nucleus' in the middle (e.g. quartz grain, piece of shell) and concentric coatings of fine-grained calcite.

X-sections:



Peloids

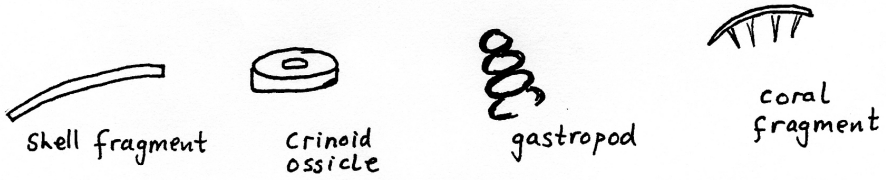
Rounded grains of structureless fine-grained carbonate.



Bioclasts

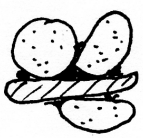
All hard parts of organisms, complete or fragmented.

e.g.



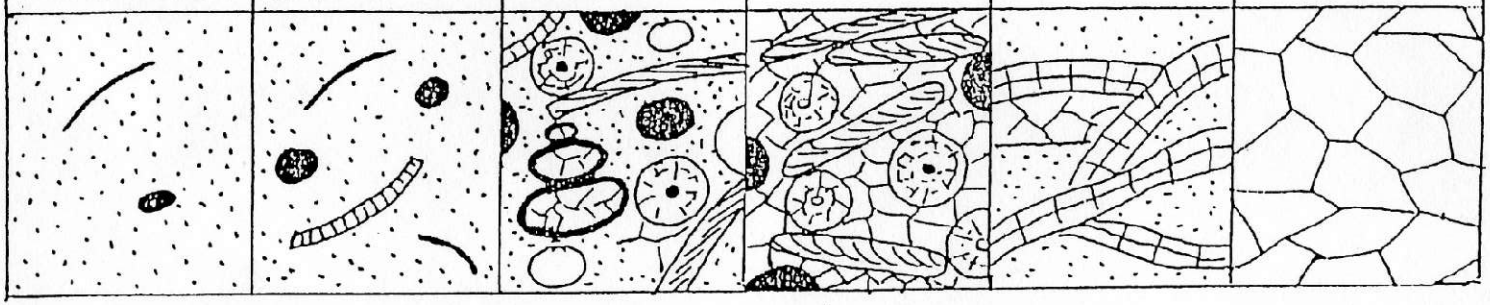
Intraclasts

Most difficult category. Literally means a re-worked fragments e.g. angular fragments of carbonate mud or aggregate grains.



original separate grains cemented together to form an aggregate grain.

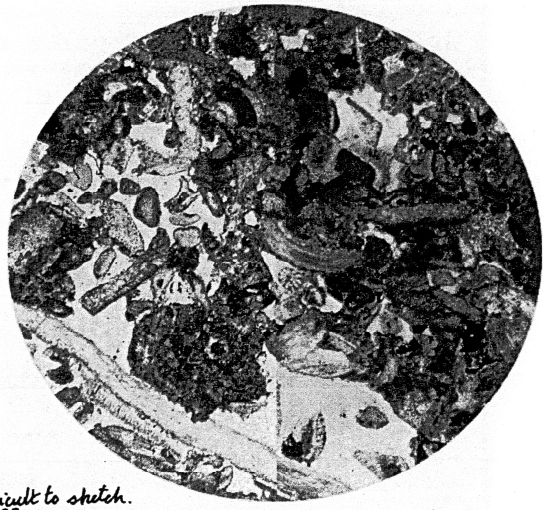
Depositional texture recognisable					Depositional texture unrecognisable
Original components not bound together during deposition				Original components bound together during deposition	
Contains carbonate mud		Grain-supported	Lacks carbonate mud. Grain-supported		
Mud-supported					
<10% grains	>10% grains				
MUDSTONE	WACKESTONE	PACKSTONE	GRAINSTONE	BOUNDSTONE	CRYSTALLINE

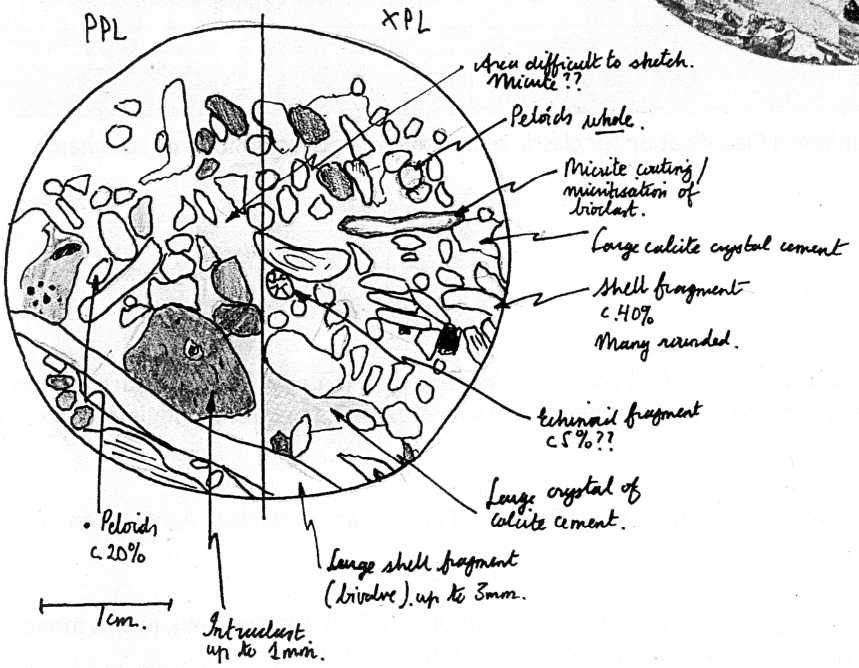


The Dunham classification of limestones, based on depositional texture.

Worked example: Thin section description

Thin section view





- GRAINS (>70%)
  - → Shell fragments 40% - fragmented, rounded, micritised, up to 3mm.
  - → Peloids 20-30% - whole, <1mm.
  - → Intraclasts - up to 2mm? 20%?
  - → Bryozoa + echinoids, fragmented, up to 1mm. 20%?

- MATRIX (<10%)
  - → Micrite between some grains only locally.

- CEMENT (c. 20%?)
  - → Large calcite crystals up to 2mm.

DESCRIPTIVE NAME : Bioclastic, peloidal packstone, with c 75% grains, 10% micrite matrix and c. 20% large crystals of calcite cement.